MARKET EARTH

SYSTEM ARCHITECTURE & TOKENOMICS SPECIFICATION

Document Version: 1.0-Alpha | **Target Network:** Internet Computer Protocol (ICP) / Base | **Domain Anchor:** me-arch.base.eth

1. Executive Summary & Vision

Market Earth is a decentralized platform architecture designed to transition economic alignment from extractive models to regenerative, collective stewardship. By utilizing a dual-token governance framework (***Me-Coin*** and ***We-Coin***), the system balances individual incentives with community-level resource coordination, leveraging immutable smart contracts on-chain.

2. Tokenomics & Dual-Token Architecture

2.1 Me-Coin (Utility & Labor Token)

- **Purpose:** Acts as the foundational unit of individual participation, acknowledging real-world or digital contribution and task execution within the ecosystem.
- **Minting Mechanics:** Minted dynamically based on validated actions, protocol engagement, or proofs of work.
- **Supply Dynamics:** Elastic supply tied directly to active platform participation, designed to prevent stagnant hoarding while ensuring fair individual representation.

2.2 We-Coin (Governance & Stewardship Token)

- **Purpose:** Drives collective coordination, voting rights, and resource allocation across public proposals.
- **Acquisition (Liquid Conversion):** Users lock up ("stake") a specific ratio of ***Me-Coins*** into the protocol contract to mint or receive corresponding ***We-Coin*** voting power.
- **Governance Utility:** Spent or committed during proposal voting cycles to ensure skin-in-the-game decision-making for ecosystem evolution.

3. Core State Machine & Smart Contract Logic (Motoko)

The backend logic operates on a canister-based state architecture managing user profiles, balances, and proposals:

```
actor MarketEarthCore {

  type User = {
    meCoins : Nat;
    weCoins : Nat;
  };

  type Proposal = {
    id : Nat;
    title : Text;
    description : Text;
    yesVotes : Nat;
    noVotes : Nat;
  };

  // State maps managed securely within the canister

  public shared (msg) func performWork() : async Nat {
    // Logic to mint Me-Coins for validated user effort
  }

  public shared (msg) func stakeForWeCoins(amount : Nat) : async Result.Result<Nat, Text>
{
    // Logic to lock Me-Coins and issue We-Coins
  }

  public shared (msg) func voteOnProposal(proposalId : Nat, support : Bool) : async
Result.Result<(), Text> {
    // Logic to deduct We-Coin governance power and record vote tally
  }
}
```

4. Infrastructure & Deployment Topology

- **Smart Contract Execution:** Internet Computer Protocol (ICP) canisters handling high-speed, low-cost decentralized computation and state storage.
- **Web3 Identity:** Integrated via Internet Identity (***id.ai***) for secure, passwordless cryptographic authentication.
- **Domain & Cross-Chain Anchors:** Registered under ***me-arch.base.eth*** to bridge decentralized identity and gateway routing across EVM-compatible networks.

5. Development Roadmap & Milestones

- **Phase 1 (Current - Q3 2026):** Core protocol architecture design, time-matrix synchronization, and sandbox prototyping using AI-assisted canister deployment tools (Caffeine/Motoko).
- **Phase 2 (Q4 2026 – Q2 2027):** Internal testnet trials, security audits, and multi-signature governance integration.

- **Phase 3 (Target: October 2027):** Public mainnet launch and activation of the decentralized autonomous framework.